Final Project Documentation – Metro Ticket Generating System in ServiceNow

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1. Introduction

Project Title: Metro Ticket Generating System in ServiceNow. Digitize metro ticket booking using ServiceNow to enable fast, accessible, and automated ticket generation through QR codes. The system allows users to book tickets by selecting stations and travel details, automatically calculates fares, and generates digital tickets instantly. It streamlines metro operations using Service Catalog, Flow Designer, and Notifications for a seamless and efficient travel experience.

2. Project Overview

Purpose: digitize ticket booking, automate fares and generate QR-based digital tickets. Features: Service Catalog item, field validations, auto-populate logic, fare logic script, QR code integration, station mapping, Flow Designer, Notifications and UI pages.

3. Architecture

UI pages and the catalog provide the experience. The fare logic script, Flow Designer and QR code integration provide the logic. Ticket and station records are stored in ServiceNow tables. Data architecture is the blueprint for how data is collected, stored, managed, integrated, and used across an organization. It defines the policies, standards, models, and rules that govern how data flows and supports business objectives.

4. Setup Instructions

Create the catalog form with variables; add the fare logic script; integrate QR code generation; map station details; build UI pages; configure Flow Designer and Notifications; run UAT and deploy.

5. Configuration Structure

Catalog item: ticket booking with Source, Destination, Passenger Type, Number of Tickets. Field validations: non-empty source/destination, fare match, approved travel date. Mandatory fields on client and server. Auto-populate defaults using catalog client scripts and UI policies. Field mapping via Process Automation Engine.

6. Running the Application

Open the booking page, select stations and travel details, submit; verify the fare, the QR digital ticket and the notification.

7. API Documentation

No custom external REST API is described; QR code generation is handled through the configured QR code integration.

8. Authentication

Handled by the ServiceNow instance and role-based access for Passengers, Station Managers, Metro Operations Team and IT Admins.

9. User Interface

Booking catalog form, digital ticket with QR code and UI pages. Insert actual ServiceNow screenshots in the final submission.

10. Testing

Field validation, fare calculation, QR generation, notifications and end-to-end booking were functionally tested, followed by UAT.

11. Screenshots / Demo

Recommended screenshots: booking form, validation messages, fare, QR ticket, notification and ticket record.

12. Known Issues

No formal load benchmark supplied. Online payment integration and WhatsApp booking are outside the implemented scope.

13. Future Enhancements

Digital payment gateway (UPI, cards, wallets), mobile app and WhatsApp booking, analytics dashboards, route optimization and gate-validation integration.